

PRAKTIKUM BASIS DATA

MODUL 09

Scripting Prosedure dan Function



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

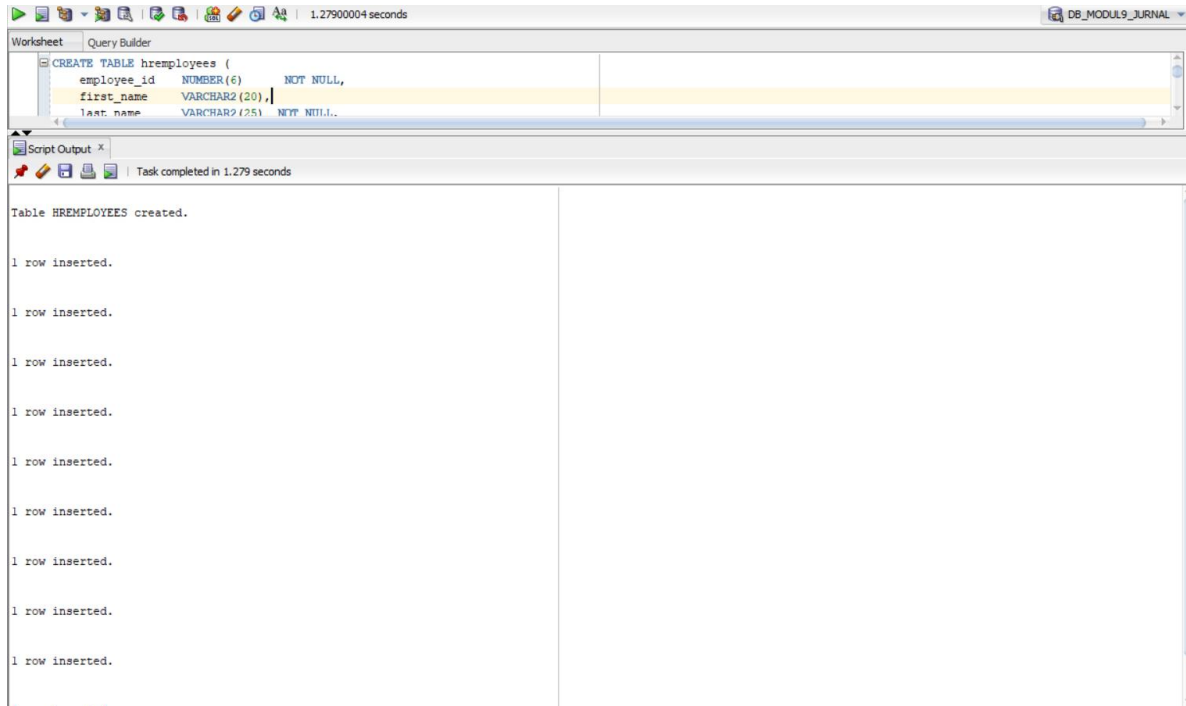
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL

Running code sql berikut, screenshot hasilnya dan analisis hasil code tersebut



1. create table employees as select * from hremployees

The screenshot shows the SQL Developer interface with a query window titled 'Query Builder'. The query contains several INSERT statements for the HREMPLOYEES table, followed by a statement to create the EMPLOYEES table as a copy of HREMPLOYEES.

```
CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)

);

-- INSERT data untuk department_id = 90 (Executive -- dipakai soal no.4)
INSERT INTO hremployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-2003','DD-MON-YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO hremployees VALUES (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005','DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO hremployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-2001','DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);

-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hremployees VALUES (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', TO_DATE('03-JAN-2006','DD-MON-YYYY'), 'IT_PROG', 9000, NULL, 102, 60);
INSERT INTO hremployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', TO_DATE('21-MAY-2007','DD-MON-YYYY'), 'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO hremployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007','DD-MON-YYYY'), 'IT_PROG', 4200, NULL, 103, 60);
INSERT INTO hremployees VALUES (124, 'Kevin', 'Mourgos', 'KMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007','DD-MON-YYYY'), 'ST_MAN', 5800, NULL, 100, 50);
INSERT INTO hremployees VALUES (141, 'Trenna', 'Rajs', 'TRAJS', '650.121.8009', TO_DATE('17-OCT-2003','DD-MON-YYYY'), 'ST_CLERK', 3500, NULL, 124, 50);
INSERT INTO hremployees VALUES (149, 'Eleni', 'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29-JAN-2008','DD-MON-YYYY'), 'SA_MAN', 10500, .2, 100, 80);
INSERT INTO hremployees VALUES (174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267', TO_DATE('11-MAY-2004','DD-MON-YYYY'), 'SA_REP', 11000, .3, 149, 80);

create table employees AS SELECT * FROM hremployees;
```

The Query Result pane shows the message: "Table EMPLOYEES created." and "Task completed in 0.152 seconds".

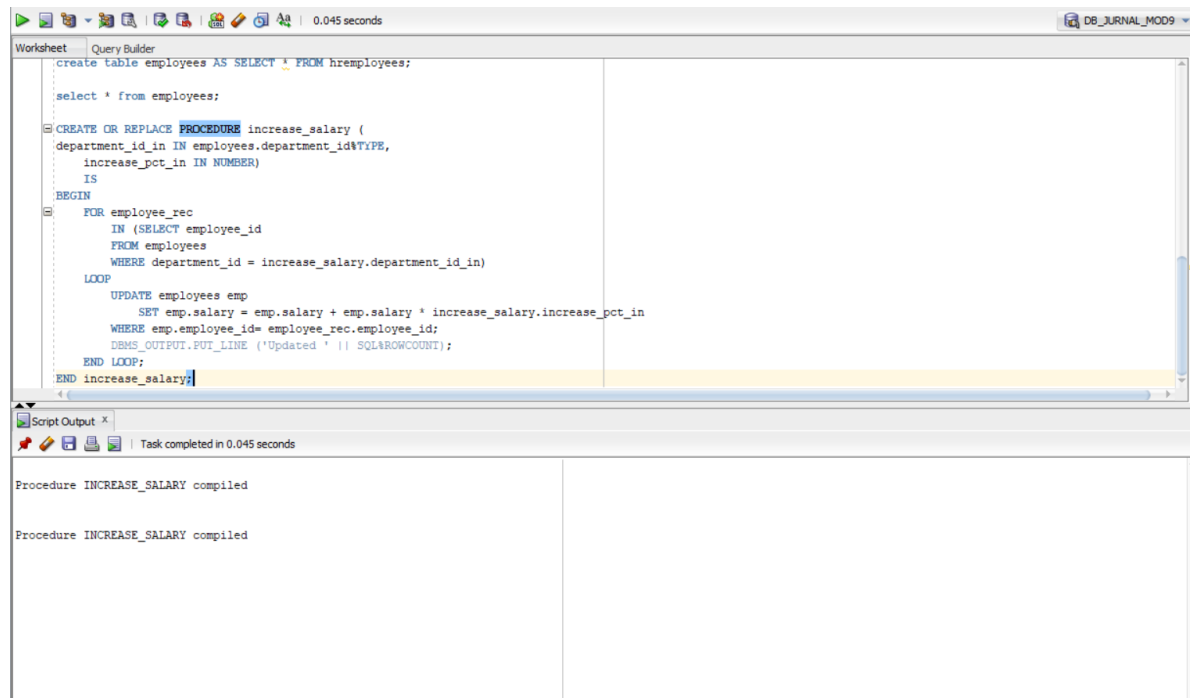
2. select * from employees

The screenshot shows the SQL Developer interface with the same query window. The query is now 'select * from employees;'. The Query Result pane displays the results of the query, showing 10 rows of employee data.

```
create table employees AS SELECT * FROM hremployees;

select * from employees;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	80



3.

4. BEGIN

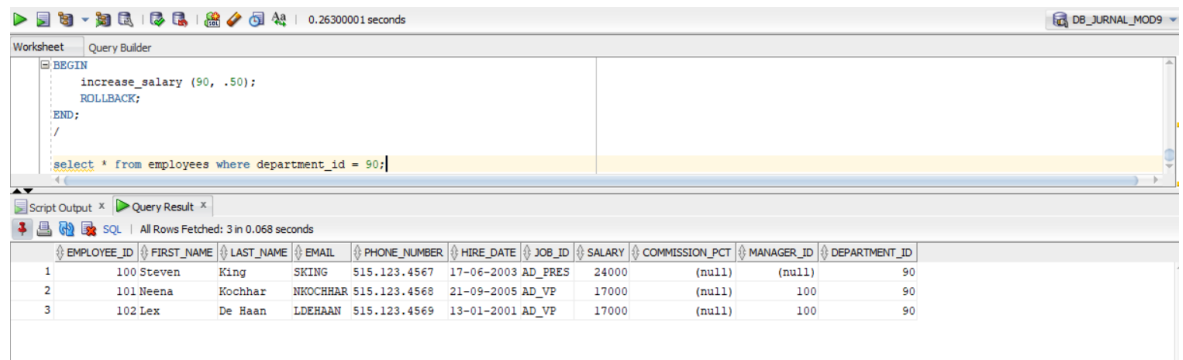
increase_salary (90, .50);

ROLLBACK;

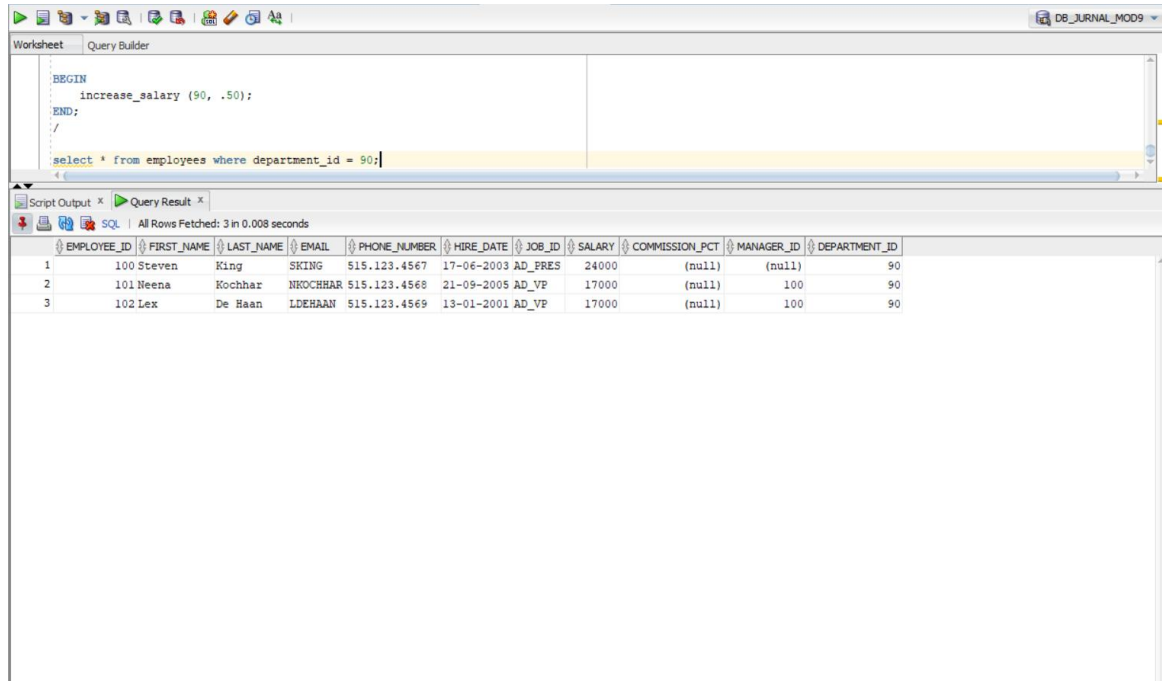
END;

/

select * from employees where department_id = 90



5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4



The screenshot shows the Oracle SQL Developer interface. The 'Script Output' tab is active, displaying the execution of a PL/SQL block. The script contains a procedure 'increase_salary' that increases salaries by 50% for employees in department 90, followed by a 'select' statement to view the results. The output shows three rows of employee data with their updated salaries.

```
BEGIN
  increase_salary (90, .50);
END;
/

select * from employees where department_id = 90;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

yang terjadi itu, jika statement ROLLBACK dihapus, maka semua perubahan yang dilakukan oleh procedure increase_salary engga akan dibatalkan. Jadi, kenaikan gajinya yang sebesar 50% untuk seluruh karyawan di department 90 akan tersimpan pada tabel employees. Setelah blok PL/SQL selesai dieksekusi, data akan tetap dalam kondisi sudah diperbarui (salary meningkat), karena tidak ada perintah untuk mengembalikannya ke kondisi semula.

~~~~~ARIGATOU GOZAIMASU~~~~~